

Bridging the Divide: A Multi-Disciplinary Circuit Breaker for Trustworthy AI in Systems Engineering

Abstract

The accelerating integration of Artificial Intelligence, particularly Large Language Models (LLMs) and Small Language Models (SLMs), into complex autonomous systems presents unprecedented opportunities and critical challenges. This paper proposes a novel "AI Circuit Breaker" framework designed to mitigate significant error propagation, hallucination, drift, and model collapse, fostering trust in autonomous agents. Drawing upon a synthesis of Systems Engineering, Control Theory, Complex Adaptive Systems, Knowledge Engineering, and advanced Metrology, this framework emphasizes a multi-layer validation logic, semantic guardrails, and robust decision modeling. It advocates for a transformative shift within organizations like INCOSE, encouraging deep interdisciplinary collaboration to architect resilient, transparent, and ultimately trustworthy AI-driven systems.

1. Introduction: The Imperative for Trustworthy AI in Complex Systems

The promise of AI to revolutionize engineering, from design optimization to autonomous operation, is undeniable. However, the probabilistic nature of modern AI, especially LLMs, introduces inherent uncertainties that can manifest as "super-huberous" assertions, leading to potentially catastrophic failures in safety-critical applications. This paper posits that to harness AI's full potential, we must first establish robust mechanisms to ensure its reliability and trustworthiness. We introduce a "Ground Truth Circuit Breaker" as a foundational element for secure AI integration, designed to operate across the entire data-to-decision pipeline of an autonomous agent.

2. The Propagation of Error: From Data Noise to Model Collapse

Error in an AI-driven autonomous system is not static; it evolves and amplifies through successive layers, as illustrated in our conceptual Circuit Breaker diagram:

- **Data Layer (Sensory Measures):** The initial point of entry for error, ranging from sensor noise and calibration inaccuracies to atmospheric interference. Without high-precision metrology ("You cannot manage what you cannot measure"), subsequent layers receive a distorted view of reality.
- **Algorithmic Layer (Deterministic Models):** Traditional algorithms process raw data. Inflexibilities or unhandled edge cases can transform sensor noise into processing artifacts or false features, which are then passed upstream as seemingly valid data points.
- **LLM Layer (Semantic Amplification & Hallucination):** LLMs, driven by probabilistic inference, interpret these processed artifacts. This is where "super-huberous" confidence

can manifest, as the LLM attempts to find semantic meaning in erroneous data, leading to convincing but false narratives (hallucinations).

- **Recursive Learning & Model Collapse:** When an autonomous agent's actions, influenced by LLM hallucinations, are fed back into its training data, a dangerous positive feedback loop emerges. The model "learns" from its own errors, reinforcing biases and gradually losing its grounding in real-world physics and logic, leading to "model collapse."
- **Autonomous Agent (Cascading Failure):** The culmination of these errors can result in an agent executing dangerous, unintended actions in the physical world, driven by a flawed internal understanding.

3. The AI Circuit Breaker Framework: A Multi-Disciplinary Solution

The proposed AI Circuit Breaker is a holistic framework built upon the synergistic application of diverse engineering disciplines, acting as a crucial intermediary between AI reasoning and physical world execution.

3.1. Core Components & Information Flow

As depicted in the accompanying illustration, the framework integrates:

- **LLM (Strategic Planning) & SLM (Tactical Perception):** These provide the AI's "intent" – high-level goals from long-range data (LLM) and rapid local interpretations from short-range data (SLM).
- **Deterministic Sensors (Physics & Proximity):** These represent the undeniable "ground truth" of the physical world, providing precise measurements and immediate feedback.
- **Circuit Breaker Logic:** The central control mechanism. It continuously evaluates the "Delta" (discrepancy) between the AI's "Intent" and the "Sensor State." If this Delta exceeds a predefined "Tolerance," the breaker "trips."

3.2. Multi-Layer Validation Logic & Brokers

The intelligence of the Circuit Breaker lies in its sophisticated validation layers, facilitated by specialized "Brokers" that enable inter-component communication and enforcement:

1. **Context Engineering & Knowledge Engineering:** These brokers provide the foundational semantic understanding.
 - **Knowledge Graph:** A structured repository of domain-specific facts, rules, and relationships. It anchors the LLM's probabilistic outputs in verifiable truths, acting as a semantic guardrail against factual hallucinations.
 - **Ontology / Semantic Guardrails:** Formally defined sets of concepts and categories that constrain the interpretations of LLMs and SLMs. They ensure semantic consistency and prevent the AI from generating impossible or

contradictory statements (e.g., a "50-foot retaining wall cannot suddenly appear in 2 seconds").

2. **Control Theory & Systems Engineering (Decision Modeling & Redundancy):** These disciplines provide the operational logic and architectural robustness.
 - **Multi-Dimensional Trust Metrics:** Beyond simple accuracy, trust is measured across dimensions like robustness, transparency, fairness, and accountability. This informs the dynamic adjustment of tolerance thresholds.
 - **Decision Modeling & Redundancy:** Multiple decision models (e.g., probabilistic, rule-based, fuzzy logic) are employed, and their outputs are compared. Significant divergence among these models can trigger the breaker, indicating potential drift. Redundant sensor arrays and processing chains provide parallel checks.
 - **Veto Power:** The deterministic sensors and their associated logic are granted ultimate "Veto Power," capable of overriding any AI-derived intent if a direct physical threat is detected.
3. **Complex Adaptive Theory (Drift, Hallucination, Model Collapse Identification):** This perspective acknowledges the dynamic and evolving nature of AI systems.
 - **Drift Detection:** Continuous monitoring of the performance characteristics (e.g., accuracy, latency) of LLMs and SLMs against baseline metrics. Significant deviations indicate model drift.
 - **Hallucination Detection:** Actively identifying discrepancies between AI-generated narratives/predictions and ground truth data, especially when conflicting with deterministic sensor readings or the Knowledge Graph.
 - **Model Collapse Identification:** Tracking the degradation of a model's performance over successive recursive learning cycles, particularly when self-generated data is incorporated without sufficient ground truth validation.

3.3. Response to Ground Truth Violation

When the Circuit Breaker trips:

- **System Halt / Safe State Activation:** The autonomous agent immediately enters a predefined safe state (e.g., emergency stop, reversion to manual control).
- **Flagging for Exclusion (Recursive Learning):** The data associated with the "Ground Truth Violation" event is explicitly flagged and excluded from future recursive learning cycles. This is critical to prevent the propagation of error and to safeguard against model collapse. The AI learns from its successes, not its "huberous" failures.

INCOSE's Role in Fostering Cross-Disciplinary Collaboration

The successful implementation of such an AI Circuit Breaker demands a profound shift towards radical interdisciplinary collaboration, a core tenet that INCOSE is uniquely positioned to champion the following.

- **Breaking Down Silos:** The framework explicitly integrates expertise from:

- **Systems Engineering:** For holistic architectural design, interface definition, and lifecycle management of complex AI-human-physical systems.
- **Software Engineering:** For robust code, secure AI development, and data integrity.
- **Control Systems Engineering:** For deterministic safety logic, stability analysis, and feedback loop design.
- **Metrology & Sensor Fusion:** For high-precision data acquisition and sensor redundancy.
- **Knowledge Engineering & Ontology Design:** For structuring semantic information and defining ground truths.
- **Cognitive Science / AI Ethics:** For understanding model biases, explainability, and the multi-dimensional aspects of trust.

Risk Management: For identifying potential failure modes and designing mitigation strategies.

- **Developing New Standards and Best Practices:** INCOSE can lead the charge in establishing new standards for AI verification, validation, and testing (VV&T) that account for probabilistic reasoning, semantic consistency, and real-time ground truth validation.
- **Curriculum Development and Training:** Promoting educational programs that emphasize the convergence of these disciplines, creating a new generation of "renaissance engineers" capable of designing trustworthy AI systems.
- **Facilitating Cross-Industry Working Groups:** Creating platforms for sharing knowledge, lessons learned, and emergent best practices in AI safety across diverse sectors (e.g., automotive, aerospace, healthcare, manufacturing).

5. Conclusion: Architecting Trust for the AI Era

The integration of advanced AI into critical systems necessitates a fundamental re-evaluation of how we design, validate, and trust technology. The "AI Circuit Breaker" framework, leveraging multi-layer validation, semantic guardrails, and robust decision modeling, provides a pragmatic pathway to mitigate the inherent risks of AI, particularly the dangers of drift, hallucination, and model collapse.

This transformative shift requires a concerted effort from all engineering disciplines, with organizations like INCOSE playing a pivotal role in fostering this essential collaboration. By collectively building these protective layers – these "Ground Truth Circuit Breakers" – we can ensure that AI serves as a powerful, reliable, and ultimately trustworthy tool for advancing human endeavors, rather than becoming a source of systemic "super-huborous" risk. The future of autonomous systems depends not just on intelligence, but on a meticulously engineered foundation of trust.